

Adhiraj Ghosh

Tübingen, Germany

[LinkedIn Profile](#)
[GitHub Profile](#)

Email : [Bethgelab](#) / [Personal](#)
Mobile : +49-17679828923(DE)

EDUCATION

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- **ELLIS / University of Tübingen / Max Planck Institute** Tübingen, Germany
PhD in Machine Learning Jul 2025 – Present
 - **University of Tübingen** Tübingen, Germany
MSc in Machine Learning Oct 2022 – Feb 2025
Thesis : Holistic Understanding of Vision-Language Models
 - **Manipal Institute of Technology / Singapore Management University** Manipal/Singapore
Bachelors in Electrical and Electronics Engineering Aug 2016 – Aug 2020
Thesis: Towards the Analysis of High Dimensional Data in Computer Vision

PUBLICATIONS [[Google Scholar](#)]

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1. Matteo Farina*, Vishaal Udandarao*, Thao Nguyen*, **Adhiraj Ghosh**, ..., et al. *DataComp-VLM: Improving Open Datasets for Vision-Language Models*. Under review, 2026.
 2. **Adhiraj Ghosh**, Vishaal Udandarao*, Thao Nguyen*, Matteo Farina*, Mehdi Cherti, Jenia Jitsev, Sewoong Oh, Elisa Ricci, Ludwig Schmidt, Matthias Bethge, **Concept-Aware Batch Sampling Improves Language-Image Pretraining**, *CVPR 2026* [[paper](#)] [[project page](#)] [[dataset](#)] [[code](#)]
 3. **Adhiraj Ghosh***, Sebastian Dziadzio*, Ameya Prabhu, Vishaal Udandarao, Samuel Albanie, Matthias Bethge, **ONEBench to Test Them All: Sample-Level Benchmarking Over Open-Ended Capabilities**, *ACL 2025* [[paper](#)] [[project page](#)]
 4. Vishaal Udandarao*, Ameya Prabhu*, **Adhiraj Ghosh**, Yash Sharma, Philip HS Torr, Adel Bibi, Samuel Albanie, Matthias Bethge, **No “Zero-Shot” Without Exponential Data: Pretraining Concept Frequency Determines Multimodal Model Performance**, *NeurIPS 2024* [[paper](#)] [[code](#)]
 5. Hassan Shahmohammadi, **Adhiraj Ghosh** and Hendrik P. A. Lensch, **ViPE: Visualise Pretty-much Everything**, *EMNLP 2023 (Outstanding Paper)* [[paper](#)] [[code](#)]
 6. **Adhiraj Ghosh**, Kuruparan Shanmugalingam and Wen-Yan Lin, **Relation Preserving Triplet Mining for Stabilising the Triplet Loss in Re-identification Systems**, *WACV 2023* [[paper](#)] [[code](#)]
 7. **Adhiraj Ghosh***, Lukas Tuggener*, Raphael Emberger*, Pascal Sager* *et al.* **Real World Music Object Recognition**, *TISMIR 2023* [[paper](#)] [[code](#)]
 8. **Adhiraj Ghosh** and Kamal Sarkar, **Irony Detection in Bengali Tweets: A New Dataset, Experimentation and Results**, *ICCIDS 2020* [[paper](#)] [[dataset](#)]

WORK EXPERIENCE

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- **Computer Vision Researcher, Zürich University of Applied Sciences** May 2021 - Aug 2022
Zürich, Switzerland
 - Worked in the **Center of Artificial Intelligence** under the supervision of **Dr. Thilo Stadelmann**.
 - Created a **Connected-Components-enabled Semantic Segmentation** network to tackle noisy labels for **Food Waste Analysis**, in collaboration with **Kitro AG**. Achieved state-of-the-art with a mean IoU score of **0.5219**.
 - Responsible for designing a novel adversarial learning system utilising discriminator-learned features for **Unsupervised Domain Adaptation** for Optical Music Recognition on the DeepScores dataset (synthetic) to real data, improving baseline results by **36%**. In collaboration with **ScorePad AG**.
 - **Research Assistant, Singapore Management University** Jan 2020 - Dec 2022
Singapore
 - Worked under the supervision of **Dr. Wen-Yan Lin** on the project - **Robust Compute-Efficient Re-Identification and Object Tracking for Surveillance Systems**.
 - Theorised a new Triplet Mining approach based on pixel-level **Image Feature Matching and Correspondence** models, termed as **Relation Preserving Triplet Mining (RPTM)**.
 - Achieved state-of-the-art results(**October 2021-Present**) on multiple public benchmarks and produced the first transferable and scalable algorithm for generalised re-identification tasks.

RESEARCH EXPERIENCE

- **Research Student, Tübingen AI Centre** Jan 2024 – Jan 2025
Supervisor: Dr. Matthias Bethge Tübingen, Germany
 - Master's Thesis student working on the **Holistic Understanding of Vision-Language Models**. Created **ONEBench**, **DataConcept** and exposed zero-shot generalization claims in foundation models (**No "Zero-shot"**).
- **Research Intern, Tübingen AI Centre** Mar 2023 – Sep 2023
Tübingen, Germany
 - Worked under the supervision of Dr. Hendrik P.A. Lensch on the project **Visualising Figurative Speech**. Created **LyricCanvas** and **ViPE**, models trained to understand and visualise figurative or non-lexical language.
- **Research Associate, Jadavpur University** Jun 2018 - Dec 2019
Supervisor: Dr. Kamal Sarkar Kolkata, India
 - Worked on **Irony Detection and Classification** in Bengali Tweets, funded by the Science and Engineering Research Board, Govt. of India. Created the first published dataset for irony detection and classification in Bengali.

INVITED TALKS

- University of Washington, 06/2026
- DatologyAI, 06/2026
- Metamorphic, 06/2026
- Voxel51, 05/2026

ACADEMIC HIGHLIGHTS AND RESPONSIBILITIES

Highlights

- ELLIS and ELLIOT PhD Scholarship, 2025.
- **Outstanding Paper Award** at EMNLP, Language Grounding to Vision, Robotics and Beyond track, 2023.
- RPTM taught in **Graduate Innovation Demonstration Course** at Shenzhen University, 2023.
- Best Undergraduate Seminar Presentation: **Implementation of Deep Learning in Medical Imaging**, 2019.
- One of four students (selection rate 1.6%) selected for a Cisco India-Manipal collaboration, 2019.

Student Mentoring

- **Andrey Khomutov**: University of Tübingen, 2026
- **Lukas Demey**: Ghent University (co-supervised with Dr. Karel D'Oosterlinck), 2025

Reviewer Responsibilities

- **Journals**: Transactions of Image Processing
- **Conferences**: CVPR 2026, ICLR 2026, NeurIPS 2023, ECCV 2022

TECHNICAL SKILLS

- **Topics of Interest** Data-centric Machine Learning, Vision and Language, Multimodal Pretraining, Evaluation and Benchmarking
 - **Languages** Python, MATLAB, Java
 - **Tools/Frameworks** Docker/Singularity, PyTorch, Tensorflow, OpenCV, Gym, ParaView, OpenCLIP
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